

Context-Window Degradation and Its Impact on Long-Document Research Synthesis

Abstract

Large language models (LLMs) now advertise context windows spanning from tens of thousands to millions of tokens, positioning them as candidate tools for synthesizing entire literatures, legal case files, or multi-report corpora in a single pass. However, a growing body of empirical work shows that nominal context length is a poor proxy for effective context utilization: model accuracy on information embedded in long inputs degrades non-uniformly, typically forming a U-shaped curve in which information near the beginning or end of the input is recalled far more reliably than information in the middle (Liu et al., 2024). This paper synthesizes evidence on the architectural, positional, and attention-based mechanisms underlying "context-window degradation," reviews benchmarking efforts (Needle-in-a-Haystack, LongBench, RULER, ∞ Bench) that quantify the phenomenon, and examines its consequences for long-document research synthesis tasks such as systematic literature review, multi-document question answering, and evidence aggregation. We survey current mitigation strategies—retrieval augmentation, hierarchical and recursive summarization, sparse and streaming attention, position-encoding extrapolation, and context compression—and evaluate their relative strengths and limitations. We conclude by identifying open research gaps, including the absence of standardized degradation metrics for real-world synthesis tasks, insufficient understanding of degradation in reasoning-augmented and multimodal long-context models, and the need for human-centered evaluation of AI-assisted evidence synthesis in high-stakes domains such as biomedical research.

Keywords: context window, long-context language models, lost-in-the-middle effect, retrieval-augmented generation, positional encoding, research synthesis, systematic review automation, attention degradation, benchmark evaluation, large language models

1. Introduction

The transformer architecture (Vaswani et al., 2017) underlies nearly all contemporary large language models, and its self-attention mechanism computes pairwise interactions across every token in an input sequence. This design confers considerable representational power but incurs quadratic computational cost in sequence length, which historically constrained practical context windows to a few thousand tokens. Over the past three years, architectural innovations—rotary position embeddings (Su et al., 2021), position-interpolation techniques, sparse and linear attention variants, and engineering advances in memory management—have enabled context windows of 32K, 128K, and even multi-million-token scale to be marketed as standard features of commercial and open-weight models.

This expansion has generated substantial interest in using LLMs for long-document research synthesis: the automated or semi-automated aggregation, comparison, and summarization of information distributed across lengthy or numerous source documents, as required in systematic literature reviews, legal discovery, policy analysis, and multi-report business intelligence. The appeal is obvious—a model that can genuinely "read" an entire corpus in one pass could, in principle, replace

much of the manual labor involved in evidence synthesis, which is estimated to require over sixty weeks and multiple research staff for a single systematic review (Cambridge Core / Research Synthesis Methods, 2025).

However, an expanding empirical literature demonstrates that the *nominal* context length supported by a model architecture diverges substantially from its *effective* context length—the portion of the input over which the model reliably attends and reasons. Liu et al. (2024) demonstrated that performance on multi-document question answering and key-value retrieval tasks forms a U-shaped curve as a function of the position of relevant information within the input: accuracy is highest when the answer-bearing content is near the start or end of the context and drops substantially—in some cases to near-random performance—when it is situated in the middle, a pattern now widely referred to as the “lost-in-the-middle” effect. This paper synthesizes the evidence on this and related degradation phenomena, examines their implications specifically for long-document research synthesis, surveys the current landscape of mitigation techniques, and identifies gaps warranting further investigation.

The remainder of this paper is organized as follows. Section 2 reviews background literature on transformer attention, positional encoding, and the empirical discovery of context-window degradation. Section 3 provides a technical discussion of the mechanisms believed to underlie degradation. Section 4 surveys current benchmarking and mitigation approaches. Section 5 discusses research challenges and limitations specific to long-document synthesis applications. Section 6 identifies research gaps and future directions, and Section 7 concludes.

2. Background and Related Work

2.1 The Transformer Attention Mechanism and Its Scaling Limits

The self-attention mechanism introduced by Vaswani et al. (2017) allows each token in a sequence to attend to every other token, computing a weighted representation based on learned query-key-value projections. Because this computation scales quadratically with sequence length in both time and memory, early transformer-based language models were restricted to contexts of 512–2048 tokens. Subsequent research pursued two broad strategies to extend usable context: (1) architectural modifications that reduce or restructure the attention computation (e.g., sparse, sliding-window, or linear attention), and (2) modifications to positional encoding schemes that allow models to generalize to sequence lengths beyond those seen during training.

Rotary Position Embedding (RoPE), introduced by Su et al. (2021) and later formalized in *Neurocomputing* (Su et al., 2024), encodes token position via a rotation matrix applied to query and key vectors, providing relative positional information that decays with token distance while nominally supporting arbitrary sequence lengths. RoPE and its variants (position interpolation, NTK-aware scaling, YaRN) underlie most modern long-context open-weight models, but empirical work has shown that naive extrapolation beyond the trained context length produces degraded performance, motivating a family of interpolation and fine-tuning techniques (Chen et al., cited in Xiong et al., 2023, *Effective Long-Context Scaling of Foundation Models*).

2.2 Empirical Discovery of the Lost-in-the-Middle Effect

Liu et al. (2024), in work published in *Transactions of the Association for Computational Linguistics* (TACL), conducted controlled experiments on multi-document question answering and synthetic

key-value retrieval tasks, systematically varying the position of the relevant document or key-value pair within an otherwise fixed-length context of distractor content. Across both open-weight and commercial closed models—including GPT-3.5-Turbo and early Claude models—they found that performance was highest when relevant information occurred at the very beginning or end of the input and degraded substantially, forming a characteristic U-shaped curve, when relevant information was placed in the middle of the context, even for models explicitly marketed as long-context capable (Liu et al., 2024). Critically, this degradation was observed even when the *total* context length was well within the model's stated maximum, indicating that the effect is not merely a truncation artifact but reflects a genuine failure of positional attention allocation.

Subsequent work extended these findings. A follow-up study on multi-hop question answering, "Lost in the Middle, and In-Between," confirmed that the positional degradation effect persists and compounds when multiple pieces of evidence distributed across a long context must be integrated rather than retrieved singly (arXiv:2412.10079). The HELMET benchmark (arXiv:2410.02694) extended lost-in-the-middle analysis to context lengths up to 128K tokens across several task types, finding that models generally prefer the most recent portions of context and continue to struggle with recalling facts placed earlier or centrally within the input.

2.3 Needle-in-a-Haystack and Successor Benchmarks

Independently of Liu et al.'s controlled multi-document QA setup, the "Needle-in-a-Haystack" (NIAH) probe—informally introduced by Kamradt (2023) and widely adopted across the field—inserts a single fact ("needle") at varying depths within a long distractor text ("haystack") and measures retrieval accuracy as a function of depth and total context length. While influential for its simplicity, NIAH has been criticized as testing only a superficial form of long-context competence: verbatim lexical retrieval rather than the reasoning, aggregation, or multi-hop tracing that realistic synthesis tasks demand.

In response, Hsieh et al. (2024) introduced RULER, a synthetic benchmark that extends NIAH along several dimensions—multiple needle types and quantities, distractor variation, multi-hop variable tracing, and aggregation tasks such as common- and frequent-word extraction. Evaluating seventeen long-context models across thirteen tasks, RULER found that although models achieve near-perfect accuracy on the vanilla NIAH retrieval test, almost all models exhibit substantial performance drops as context length increases, and only a minority of models claiming 32K-token context sizes maintain satisfactory performance at that length (Hsieh et al., 2024). This finding is central to the present paper's thesis: retrieval-style benchmarks systematically overstate a model's effective synthesis capability relative to more demanding aggregation and reasoning tasks.

Complementing RULER, Bai et al. (2024) introduced LongBench, "the first bilingual, multi-task benchmark for long context understanding," comprising 21 datasets across six task categories—including single- and multi-document question answering, summarization, few-shot learning, synthetic tasks, and code completion—in both English and Chinese, with average input lengths of 6,711 words (English, Bai et al., 2024). LongBench evaluations similarly report that context-compression and retrieval techniques improve performance for models with otherwise weak long-context ability, but that such hybrid approaches still lag behind models with genuinely strong long-context comprehension (Bai et al., 2024).

2.4 Retrieval-Augmented Generation as a Complementary Paradigm

A parallel research thread, motivated by the limitations of parametric memory rather than context length per se, is retrieval-augmented generation (RAG). Lewis et al. (2020), in work presented at NeurIPS, introduced RAG models that combine a pretrained sequence-to-sequence generator with a non-parametric memory accessed via a learned dense retriever over a corpus such as Wikipedia, allowing the model to condition generation on retrieved passages rather than requiring all relevant information to reside within a single context window. RAG architectures set new state-of-the-art results on knowledge-intensive NLP tasks at the time of publication (Lewis et al., 2020) and remain foundational to current long-document synthesis pipelines, as discussed in Section 4.

3. Main Thematic and Technical Discussion: Mechanisms of Context-Window Degradation

Context-window degradation is not a single phenomenon but an emergent consequence of several interacting architectural and training factors.

3.1 Positional Encoding Extrapolation Failure

Because most transformer position-encoding schemes—including learned absolute embeddings and, to a lesser but still measurable extent, RoPE—are trained on sequences up to a fixed maximum length, models frequently exhibit degraded attention behavior when applied to positions far outside the training distribution, even when architecturally capable of accepting longer inputs. Research on RoPE extrapolation (e.g., work building on Su et al., 2021 and Xiong et al., 2023) shows that naive application beyond the trained length produces distorted attention distributions, motivating interpolation-based scaling methods; even with such methods, degradation at extreme lengths is reduced but not eliminated. Related work on numerical precision (arXiv:2411.13476) shows that low-precision (bfloat16) implementations of RoPE introduce additional positional encoding artifacts that compound with sequence length, an underappreciated engineering contributor to degradation.

3.2 Attention Dilution and the U-Shaped Recency/Primacy Bias

As context length grows, the attention mass available to any individual token, when normalized across a softmax over an increasingly large key set, is diluted—each token competes with a much larger pool of alternatives for attention weight. Liu et al. (2024) interpret their U-shaped findings as consistent with models having learned strong primacy and recency biases during training (where relevant information is disproportionately near the beginning of prompts or immediately preceding the point of generation), biases that transfer poorly to genuinely long, information-dense inputs where relevance is distributed arbitrarily. A related survey on model robustness (arXiv:2503.23924) confirms that this primacy/recency bias generalizes across model families, including GPT-3.5-Turbo and Claude, and interacts with task type—aggregation and multi-hop reasoning tasks in RULER and ∞ Bench show even sharper degradation than pure retrieval (arXiv:2503.23924).

3.3 The Attention Sink Phenomenon

Xiao et al. (2024), presented at ICLR, identified a distinct but related phenomenon termed “attention sink,” in which initial tokens in a sequence receive disproportionately high attention scores irrespective of their semantic relevance, apparently because the softmax normalization requires attention mass to be allocated somewhere, and early tokens—visible to all subsequent positions—

become a default sink. This finding motivated StreamingLLM, a KV-cache management framework that preserves the attention-sink tokens alongside a sliding window of recent tokens, enabling stable generation over sequences of up to four million tokens without fine-tuning (Xiao et al., 2024). While primarily an efficiency solution, the attention-sink phenomenon has direct relevance to context-window degradation: it demonstrates that attention allocation in long contexts is governed partly by structural artifacts of the softmax mechanism rather than purely by content relevance, reinforcing why centrally located, semantically important information can be systematically under-attended.

3.4 Task-Dependent Severity: Retrieval versus Aggregation versus Reasoning

The severity of degradation is highly task-dependent. RULER's evaluation across retrieval, multi-hop tracing, and aggregation task categories found that while models can approach ceiling performance on simple single-needle retrieval even at long lengths, performance on multi-hop variable tracing and aggregation tasks—both closer analogues to genuine research synthesis, which requires integrating and comparing information across sources rather than locating a single fact—declines far more steeply as context length increases (Hsieh et al., 2024). This distinction is critical for research synthesis applications: a model's strong score on a needle-retrieval benchmark provides little assurance of its ability to reliably aggregate claims, resolve conflicting evidence, or trace multi-hop dependencies across a genuinely long document set.

3.5 Compounding Effects in Multi-Document and Multi-Hop Settings

Follow-up work on multi-hop question answering explicitly extending the “lost in the middle” framework found that when a correct answer requires synthesizing evidence from two or more documents positioned at different depths within the context, degradation compounds relative to single-document retrieval; misplacement of any one of the required evidence documents toward the middle of the context reduces overall accuracy substantially (arXiv:2412.10079). Because most realistic long-document research synthesis tasks—systematic reviews, cross-report comparison, multi-source policy analysis—require exactly this kind of multi-hop, multi-document integration rather than single-fact retrieval, this compounding effect suggests that real-world degradation in synthesis applications may be considerably more severe than single-needle benchmarks indicate.

4. Current Approaches and Developments

4.1 Retrieval-Augmented Generation and Hybrid Retrieval-Context Pipelines

The most widely deployed mitigation for context-window degradation in applied research-synthesis systems is retrieval augmentation. Rather than presenting an entire document corpus within a single context window, RAG pipelines (building on the paradigm established by Lewis et al., 2020) retrieve only the passages estimated to be most relevant to a given query, substantially reducing the effective context length the generator must reason over and, in principle, sidestepping positional degradation by ensuring relevant content is concentrated rather than diffusely distributed. LongBench's evaluations found that context-compression and retrieval techniques improve performance specifically for models with otherwise weak native long-context ability, though hybrid pipelines still underperform models with strong intrinsic long-context comprehension on tasks requiring integration across many passages (Bai et al., 2024). RAG's limitations for synthesis tasks include retriever recall failures (relevant passages not surfaced at all), loss of cross-document

context that spans retrieved chunk boundaries, and the difficulty of using dense retrieval effectively for aggregation queries where “relevance” is diffuse across an entire corpus rather than concentrated in a few passages.

4.2 Hierarchical and Recursive Summarization

An alternative to retrieval is hierarchical decomposition: dividing a long document or corpus into segments, summarizing each independently, and recursively summarizing the resulting summaries until the content fits within a manageable context window. Recursive abstractive approaches such as RAPTOR (cited in recent RAG literature, e.g., arXiv:2510.10114) construct a tree of recursively generated summaries over document chunks, enabling retrieval and synthesis at multiple levels of abstraction. This approach reduces the positional dilution problem by never requiring a single attention pass over the full, undivided source material, but introduces its own risk: information loss or distortion accumulates at each summarization stage, and errors or omissions in early-stage summaries propagate to later synthesis, a concern particularly salient in evidence-sensitive domains such as biomedical or legal research synthesis.

4.3 Sparse, Streaming, and Efficient Attention Mechanisms

A substantial body of systems-oriented research addresses degradation and computational cost jointly through modified attention mechanisms. StreamingLLM (Xiao et al., 2024) preserves attention-sink tokens alongside a sliding window to maintain stable generation over effectively unbounded sequence lengths, achieving up to a 22.2-times speedup over naive sliding-window recomputation (Xiao et al., 2024). Related KV-cache compression and query-aware sparsity methods (e.g., H2O, Quest, DuoAttention, surveyed in arXiv:2508.09834) selectively retain only the most contextually important key-value pairs, reducing memory overhead while attempting to preserve accuracy. However, these methods are primarily designed to extend the *feasible* context length under fixed memory budgets rather than to correct the underlying positional attention bias documented by Liu et al. (2024); a sparse-attention model can still exhibit lost-in-the-middle behavior over the tokens it does retain.

4.4 Position-Encoding Extension and Long-Context Fine-Tuning

Position interpolation, NTK-aware RoPE scaling, and YaRN-style methods rescale or reinterpret positional encodings to allow models trained at shorter context lengths to generalize to longer ones, often combined with continued pretraining or fine-tuning on long-sequence data (Xiong et al., 2023). Such approaches have measurably extended the *usable* context length of open-weight models but have not eliminated the U-shaped degradation curve documented across benchmarks; RULER’s finding that only a subset of models claiming 32K-token support maintain satisfactory performance at that length (Hsieh et al., 2024) applies to models employing these extension techniques as much as to those without them, indicating that context-length extension and effective-context reliability are only loosely coupled engineering problems.

4.5 LLM-Assisted Evidence Synthesis in Practice

Outside of benchmark settings, applied research has begun evaluating LLM performance on realistic research-synthesis subtasks, most extensively in biomedical systematic review screening. Multiple validation studies report that LLMs such as GPT-4o, GPT-4-Turbo, and compact open-weight models (Llama 3.1 8B, Gemma 2 9B) can achieve high sensitivity—in some studies approaching 100%—for title-and-abstract screening against human-curated systematic review inclusion decisions, with

substantial workload reduction, though often at the cost of lower precision, requiring human arbitration of borderline cases (Cambridge Core / Research Synthesis Methods, 2025; medRxiv, 2024). Notably, these screening tasks operate over comparatively short text units (titles and abstracts) rather than full-length documents, and thus largely avoid triggering severe context-window degradation; the more demanding step of full-text synthesis and cross-study aggregation—precisely where lost-in-the-middle effects would be expected to bind most tightly—remains comparatively under-evaluated in this applied literature, a gap discussed further in Section 6.

5. Research Challenges and Limitations

Divergence between benchmark and real-world document structure. Most quantitative evidence on context-window degradation derives from synthetic or semi-synthetic benchmarks (NIAH, RULER, key-value retrieval) or curated multi-document QA datasets (Liu et al., 2024; Hsieh et al., 2024). These benchmarks intentionally control for confounds but may not faithfully represent the structure of real research documents—which contain redundancy, cross-referencing, heterogeneous formatting, tables, and citations—raising questions about the external validity of degradation estimates for genuine long-document synthesis workflows.

Confounding between context length and task difficulty. Because longer contexts in realistic settings often also contain more distinct claims, more potential contradictions, and greater topical heterogeneity, it is difficult to cleanly separate degradation attributable to positional/attentional mechanisms from degradation attributable to genuinely higher task complexity. RULER's aggregation and multi-hop tasks partially address this by holding task structure fixed while varying length (Hsieh et al., 2024), but applied research-synthesis studies rarely offer this level of experimental control.

Precision and engineering artifacts compound architectural limitations. Findings that low-precision arithmetic (bfloat16) can itself distort RoPE-based positional encoding at long sequence lengths (arXiv:2411.13476) indicate that degradation observed in deployed systems reflects a combination of fundamental architectural bias and lower-level implementation choices, complicating efforts to attribute performance drops to any single cause and complicating cross-study comparability when papers do not report numerical precision.

Retrieval and compression introduce new failure modes rather than eliminating old ones. RAG-based mitigation (Section 4.1) reduces the effective context a generator must process but is bounded by retriever recall; a synthesis system that never retrieves a relevant passage will fail regardless of the generator's context-handling ability. Hierarchical summarization (Section 4.2) similarly risks compounding information loss across summarization stages. Neither approach has been rigorously benchmarked against full-context baselines specifically on aggregation-style research-synthesis tasks at scale.

Insufficient evaluation of full-text, cross-study synthesis. As noted in Section 4.5, the strongest applied evidence for LLM-assisted evidence synthesis concerns abstract-level screening, a comparatively short-context task. Evidence on LLM reliability for full-text extraction and cross-study qualitative synthesis—the stages most exposed to context-window degradation—is considerably sparser, and existing abstract-screening accuracy figures should not be extrapolated to full-document synthesis performance.

Domain-specific risk in high-stakes synthesis. In domains such as biomedicine, law, and policy, an LLM’s silent omission of centrally located but critical evidence (a plausible consequence of lost-in-the-middle degradation) is a qualitatively different and potentially more dangerous failure mode than an obvious refusal or explicit uncertainty flag, because the omission is not visibly signaled to the human reviewer relying on the synthesis output. Current benchmarks generally report aggregate accuracy rather than characterizing the downstream risk profile of silent omission errors.

6. Research Gaps and Future Directions

Standardized degradation metrics for applied synthesis tasks. Existing benchmarks (NIAH, RULER, LongBench, HELMET) provide valuable but largely retrieval- or QA-centric measures of long-context competence. There is a research gap in standardized, position-aware evaluation protocols specifically designed around research-synthesis operations—claim aggregation, contradiction detection, and cross-source attribution—rather than single-answer retrieval, that would allow direct comparison of degradation severity across models for synthesis-relevant capabilities.

Full-text and multi-document synthesis benchmarks in applied domains. Given the concentration of applied validation studies on abstract-level screening (Section 4.5), future work should extend rigorous diagnostic-accuracy methodology to full-text extraction and multi-study qualitative synthesis, explicitly varying the position of critical evidence within long documents to test for lost-in-the-middle effects in realistic biomedical, legal, and policy corpora.

Mechanistic interpretability of positional attention bias. While the attention-sink phenomenon (Xiao et al., 2024) and U-shaped recall curves (Liu et al., 2024) are well documented empirically, a more complete mechanistic account of why models develop primacy/recency biases during training—and whether targeted training interventions (e.g., data augmentation with relevant information at varied positions) can durably correct rather than merely mask the bias—remains underdeveloped.

Interaction between context-window degradation and emerging long-context reasoning/agentic models. Recent model families increasingly combine long context windows with extended chain-of-thought or agentic multi-step reasoning. Whether such reasoning processes are equally susceptible to positional degradation when reasoning traces themselves grow long, or whether iterative reasoning provides an implicit mitigation by allowing models to revisit and re-attend to context, is an open empirical question not yet addressed by the reviewed literature.

Human-centered evaluation and trust calibration. Because silent omission of centrally positioned evidence is a distinctive risk of context-window degradation in high-stakes synthesis, future research should examine how human reviewers using LLM-assisted synthesis tools calibrate trust and detect omissions, and whether interface designs (e.g., mandatory source attribution, confidence flags tied to evidence position) can mitigate downstream harm even when the underlying degradation is not fully resolved architecturally.

Standardizing engineering reporting. Given evidence that numerical precision alone can measurably affect positional encoding fidelity at long sequence lengths (arXiv:2411.13476), the field would benefit from standardized reporting of precision, position-encoding scheme, and context-extension method alongside degradation benchmark results, to enable more reliable cross-study and cross-model comparison.

7. Conclusion

The expansion of nominal context windows in large language models has not been matched by a proportional expansion of effective, reliable attention over long inputs. Converging evidence from controlled multi-document question-answering studies (Liu et al., 2024), synthetic retrieval-and-aggregation benchmarks (Hsieh et al., 2024), bilingual multitask evaluation (Bai et al., 2024), and mechanistic analyses of attention allocation (Xiao et al., 2024) demonstrates that context-window degradation is a robust, multi-causal phenomenon spanning positional encoding extrapolation limits, attention dilution, and structural attention-sink artifacts. For long-document research synthesis specifically—a use case that inherently demands integration of centrally distributed, multi-hop evidence rather than single-fact retrieval—this degradation poses a substantive and, based on current evidence, only partially mitigated risk. Retrieval augmentation, hierarchical summarization, sparse/streaming attention, and position-encoding extension each address particular facets of the problem but introduce their own limitations and have not been comprehensively validated against full-context baselines on realistic synthesis tasks. Progress in applying LLMs reliably to systematic review, legal discovery, and comparable long-document synthesis workflows will depend on closing the gap between benchmark-oriented measures of long-context competence and task-specific, position-aware evaluation of aggregation and cross-document reasoning, alongside continued mechanistic and engineering-level investigation of why relevant information “in the middle” continues to be lost.

Complete References

- Bai, Y., Lv, X., Zhang, J., Lyu, H., Tang, J., Huang, Z., Du, Z., Liu, X., Zeng, A., Hou, L., Dong, Y., Tang, J., & Li, J. (2024). LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 3119–3137). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.acl-long.172> (arXiv preprint: arXiv:2308.14508, <https://doi.org/10.48550/arXiv.2308.14508>)
- Hsieh, C.-P., Sun, S., Krizan, S., Acharya, S., Rekesh, D., Jia, F., Zhang, Y., & Ginsburg, B. (2024). RULER: What's the real context size of your long-context language models? *arXiv preprint*, arXiv:2404.06654. <https://doi.org/10.48550/arXiv.2404.06654>
- Kamradt, G. (2023). *Needle in a haystack: Pressure testing LLMs* [Software repository]. GitHub. https://github.com/gkamradt/LLMTest_NeedleInAHaystack
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. Curran Associates, Inc. arXiv preprint: arXiv:2005.11401, <https://doi.org/10.48550/arXiv.2005.11401>
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12, 157–173. https://doi.org/10.1162/tacl_a_00638
- Su, J., Lu, Y., Pan, S., Wen, B., & Liu, Y. (2021). RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint*, arXiv:2104.09864. <https://doi.org/10.48550/arXiv.2104.09864>
- Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., & Liu, Y. (2024). RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568, Article 127063.

<https://doi.org/10.1016/j.neucom.2023.127063>

Xiao, G., Tian, Y., Chen, B., Han, S., & Lewis, M. (2024). Efficient streaming language models with attention sinks. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024)*. arXiv preprint: arXiv:2309.17453, <https://doi.org/10.48550/arXiv.2309.17453>

Xiong, W., Liu, J., Molybog, I., Zhang, H., Bhargava, P., Hou, R., Martin, L., Rungta, R., Sankararaman, K. A., Oguz, B., Khabisa, M., Fang, H., Mehdad, Y., Narang, S., Malik, K., Fan, A., Bhosale, S., Edunov, S., Lewis, M., Wang, S., & Ma, H. (2023). Effective long-context scaling of foundation models. *arXiv preprint*, arXiv:2309.16039. <https://doi.org/10.48550/arXiv.2309.16039>

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*. Curran Associates, Inc. arXiv preprint: arXiv:1706.03762, <https://doi.org/10.48550/arXiv.1706.03762>

Yun, J. et al. (2024). Lost in the middle, and in-between: Enhancing language models' ability to reason over long contexts in multi-hop QA. *arXiv preprint*, arXiv:2412.10079.

<https://doi.org/10.48550/arXiv.2412.10079>

Yen, H. et al. (2024). HELMET: How to evaluate long-context language models effectively and thoroughly. *arXiv preprint*, arXiv:2410.02694. <https://doi.org/10.48550/arXiv.2410.02694>

(2024). Compact large language models for title and abstract screening in systematic reviews: An assessment of feasibility, accuracy, and workload reduction. *Research Synthesis Methods*. Cambridge University Press. <https://www.cambridge.org/core/journals/research-synthesis-methods/article/compact-large-language-models-for-title-and-abstract-screening-in-systematic-reviews-an-assessment-of-feasibility-accuracy-and-workload-reduction/CB00FD70434780029EF6C027055331BA>

(2024). Large language models for abstract screening in systematic- and scoping reviews: A diagnostic test accuracy study. *medRxiv preprint*. <https://doi.org/10.1101/2024.10.01.24314702>

(2025). Validation of large language models (Llama 3 and ChatGPT-4o mini) for title and abstract screening in biomedical systematic reviews. *Research Synthesis Methods*. Cambridge University Press. <https://www.cambridge.org/core/journals/research-synthesis-methods/article/validation-of-large-language-models-llama-3-and-chatgpt4o-mini-for-title-and-abstract-screening-in-biomedical-systematic-reviews/EDE7C95374C7FF6200B7280D5742D906>

Model Hemorrhage and the Robustness Limits of Large Language Models. (2025). *arXiv preprint*, arXiv:2503.23924. <https://doi.org/10.48550/arXiv.2503.23924>

When Precision Meets Position: BFloat16 Breaks Down RoPE in Long-Context Training. (2024). *arXiv preprint*, arXiv:2411.13476. <https://doi.org/10.48550/arXiv.2411.13476>

Speed Always Wins: A Survey on Efficient Architectures for Large Language Models. (2025). *arXiv preprint*, arXiv:2508.09834. <https://doi.org/10.48550/arXiv.2508.09834>

Note on sourcing: All references above correspond to identifiable, publicly indexed scholarly or preprint publications (ACL Anthology, TACL/MIT Press, NeurIPS Proceedings, ICLR OpenReview, arXiv, and Cambridge University Press journals) current as of the time of writing. Several 2024–2025 arXiv preprints listed here (e.g., arXiv:2412.10079, arXiv:2410.02694, arXiv:2503.23924, arXiv:2411.13476, arXiv:2508.09834) are cited by title and identifier; readers should consult the arXiv record directly via the DOI for the most current author list and any subsequent peer-reviewed publication venue, as preprints are occasionally updated or formally published after initial posting.